

EEG-based Multiple Neurodevelopmental Disorders Detection

B.Tech. Project — complete record of work, results and plan

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Contents

1. The project in one page
2. Background: problem, paper and datasets
3. The pipeline, step by step
4. What was built and fixed
5. Results: reproduction
6. Results: subject-independent evaluation and diagnostics
7. What the results mean
8. Caveats and things to be ready for
9. Mid-semester report status
10. Plan until the final evaluation
11. How to re-run everything
12. Summary in simple words

1. The project in one page

Goal. Detect two childhood neurodevelopmental disorders — intellectual and developmental disorder (IDD) and attention-deficit/hyperactivity disorder (ADHD) — from EEG, and find out how trustworthy such detectors really are.

Starting point. U. Chandela, K. N. Faisal and R. R. Sharma, “Electroencephalogram-Based Unified Approach for Multiple Neurodevelopmental Disorders Detection in Children Using Successive Multivariate Variational Mode Decomposition,” *IEEE Trans. Cogn. Develop. Syst.*, vol. 17, no. 6, pp. 1350–1359, Dec. 2025. One pipeline (SMVMD → 9 features → energy weighting → mRMR → KNN) is used for both disorders.

What makes this project different. The paper evaluates by shuffling short EEG windows into cross-validation folds, so windows from the same child appear in both training and testing. This project reproduces the pipeline and then tests it the way it would be used in practice — on children it has never seen.

Headline result.

Setting	Seen children (segment-wise)	New children (subject-wise)	Children correct
DS-1 Rest	99.69%	62.73%	10/14
DS-1 Music	96.43%	43.01%	8/14
DS-1 Rest+Music	99.77%	62.34%	9/14
DS-2 ADHD	85.81%	57.54%	74/121

Why. The features mainly identify *which child* a window came from: a nearest-neighbour test names the child with 98.91% accuracy on DS-1 rest (chance 7.14%). A classifier that has seen a child's other windows simply recognises the child.

Next. Build and validate a model that works on new children: nested subject-wise evaluation, an ablation study of the decomposition, interpretability, cross-dataset analysis and channel reduction, finishing at the end of November.

2. Background: problem, paper and datasets

2.1 Why EEG for IDD and ADHD

Both disorders start in childhood, and support started early tends to help more. Diagnosis today relies on clinical interviews and rating scales, which need trained specialists and depend on observer judgement. EEG records brain electrical activity from the scalp: it is non-invasive, cheap, tolerated well by children, and portable headsets exist. The difficulty is that EEG is noisy, changes over time and differs a lot between people, which is why signal decomposition and machine learning are used.

2.2 The two datasets

	DS-1 (IDD)	DS-2 (ADHD)
Source	Sareen et al., Data in Brief, 2020 (Mendeley)	Motie Nasrabadi et al., IEEE DataPort, 2020
Children	7 IDD + 7 typical (TDC)	61 ADHD + 60 controls (NC)
Recording	14-channel EMOTIV EPOC+, 128 Hz; 2 min rest and 2 min music	19-channel 10–20 EEG, 128 Hz; visual attention task
Windows (5 s, 50% overlap)	644 per condition (322 IDD / 322 TDC); 1288 rest+music	6588 (3680 ADHD / 2908 NC)
Windows per child	46 per condition	23–134 (recording length varies)
Note	Very small cohort: each child is 1/14 of the data	ADHD children were on methylphenidate (Ritalin); controls were not

The window counts produced by this implementation equal those stated in the paper (1288 for DS-1 and 6588 for DS-2), which confirms that loading and segmentation were reproduced correctly.

3. The pipeline, step by step

Step	What happens	Settings used
1. Preprocessing	Clean the raw EEG.	DS-1: the dataset's own cleaned files. DS-2: Butterworth 0.5–60 Hz, 50 Hz notch, Coiflet-3 wavelet + SURE denoising (level 6).
2. Windowing	Cut each recording into short pieces; each piece is one sample.	5 s windows (640 samples), 50% overlap (step 320).
3. SMVMD	Split each multichannel window into oscillatory “modes” (like separating a chord into notes). The number of modes K is chosen automatically and differs per window.	$\alpha = 2000$ (DS-1) / 1000 (DS-2), $\tau = 0$, $\text{tol} = 1e-10$; stop when residual energy $< 1\%$ or $K = 20$.
4. Features	Describe every mode on every channel with 9 numbers.	STD, VAR, RMS, IQR, ASSR, average power, max fractal length, information potential, parametric-centred correntropy.
5. Energy weighting	Combine the modes of each channel into one fixed-length vector, weighting each mode by its energy (paper Eq. 7).	$14 \times 9 = 126$ features (DS-1); $19 \times 9 = 171$ (DS-2).
6. mRMR	Rank features: most informative, least redundant first; keep the top ones.	Keep 80 (rest), 47 (music), 103 (rest+music), 171 (DS-2).
7. KNN	Classify a window by its nearest neighbours.	Hyperparameters from the paper's Table II (see Section 5). The paper's Bayesian optimisation was not re-run.
8. Evaluation	Measure accuracy with cross-validation.	Segment-wise 10-fold (paper) and subject-wise (this project): LOSO for DS-1, grouped 10-fold for DS-2.

4. What was built and fixed

4.1 Code base

Part	Contents
src/	Data loaders, preprocessing, segmentation, SMVMD, features, energy integration, mRMR, KNN, evaluation, result store (results_io.py), shared DS-1 driver (pipeline.py).
experiments/	ds1_rest, ds1_music, ds1_rest_music, ds2_adhd (the four reproductions); stage1_analysis (subject-wise protocols and diagnostics); make_report.
results/metrics/	One JSON per experiment with every metric, settings and timings.
results/features/	Cached feature matrices, so re-runs skip SMVMD (seconds instead of an hour).
results/stage1/	results_summary.json (single source of all numbers), figures, per-child CSVs.
report/	MSE report in LaTeX (IITI title page), table/number generator, checker, this record.

4.2 Problems found and fixed

Problem	Effect	Fix
DS-2 crashed on its first subject (an unexpected config argument)	No DS-2 results could ever be produced	Preprocessing accepts the argument and rejects anything but SURE
PCC feature summed only 50 time lags but divided as if it summed all	PCC values wrong ($\approx 75\%$ too high in a test)	Exact closed form over all lags, verified against brute force; also faster
Report writer crashed on Windows (text encoding)	DS-1 rest never saved results	UTF-8 output
Only one DS-1 experiment saved results; one used a wrong key for paper targets	Results lost / comparison silently empty	Shared DS-1 driver; every run saves a full bundle
Module log messages were silently discarded	Long runs looked frozen	Logging attached to the root logger
Full mRMR ranking computed and thrown away; mRMR run even when keeping all features	Hours of wasted computation	Removed; skipped when all features are kept
DS-2 mRMR redundancy matrix estimated from all 6588 windows	≈ 2.5 h for one ranking	Estimated from a seeded 2000-window sample (DS-1 unaffected: fewer windows than that)
Raw DS-2 cache saved subject IDs in a format that could not be reloaded	Raw baseline evaluation crashed	Saved as plain strings; no recomputation needed

5. Results: reproduction (paper-style evaluation)

Segment-wise stratified 10-fold cross-validation, with scaling and mRMR fitted inside each training fold. Precision and recall are class-weighted.

Setting	Windows	Features	Accuracy	Precision	Recall	F1	KNN (paper Table II)
DS-1 Rest	644	80	99.69%	99.69%	99.69%	99.69%	k=2, cityblock, uniform, standardised
DS-1 Music	644	47	96.43%	96.43%	96.43%	96.43%	k=2, cityblock, squared inverse
DS-1 Rest+Music	1288	103	99.77%	99.77%	99.77%	99.77%	k=1, euclidean, squared inverse, standardised
DS-2 ADHD	6588	171	85.81%	85.79%	85.81%	85.79%	k=2, cosine, squared inverse

DS-2 extra metrics: MCC 0.712, Cohen's κ 0.712, G-mean 0.855.

DS-2 classifier comparison (same features, segment-wise 10-fold):

Classifier	Accuracy	F1	Time per window
DT	70.04%	70.07%	0.516 ms
KNN	85.81%	85.79%	0.162 ms
SVM	65.29%	61.48%	23.630 ms
Ensemble	80.69%	80.42%	34.250 ms

mRMR feature-count sweep (accuracy as more ranked features are added):

Setting	Features kept (paper)	Best count found here
DS-1 Rest	80	70
DS-1 Music	47	100
DS-1 Rest+Music	103	95
DS-2 ADHD	171	165

6. Results: subject-independent evaluation and diagnostics

Every analysis below uses the same features and the same classifier settings; only the way the data is split, or one component, changes. Scaling, mRMR and KNN are always fitted inside each training fold.

6.1 Seen children vs new children

Setting	Overlap	Segment-wise	Subject-wise	Subject-wise bal. acc.	Children correct (vote)
DS-1 Rest	50%	99.69%	62.73%	62.73%	10/14
DS-1 Rest	0%	99.07%	63.66%	63.66%	9/14
DS-1 Music	50%	96.43%	43.01%	43.01%	8/14
DS-1 Music	0%	94.41%	50.62%	50.62%	8/14
DS-1 Rest+Music	50%	99.77%	62.34%	62.34%	9/14
DS-1 Rest+Music	0%	99.22%	63.51%	63.51%	10/14
DS-2 ADHD	50%	85.81%	57.54%	57.09%	74/121
DS-2 ADHD	0%	74.86%	58.04%	57.53%	79/121

Subject-wise = leave-one-subject-out (14 folds) for DS-1 and grouped 10-fold for DS-2: all windows of a child are held out together. "Children correct" assigns each held-out child the majority label of its windows. 0%-overlap results use the even-numbered windows only (the window is exactly twice the step), taken from the cache.

6.2 Does fitting mRMR on all data inflate accuracy?

The paper ranks features once on all windows before cross-validation. Doing the same here (segment-wise):

Setting	mRMR inside each fold	mRMR on all data (paper order)
DS-1 Rest	99.69%	99.69%
DS-1 Music	96.43%	94.10%
DS-1 Rest+Music	99.77%	99.77%

Equal or lower, so feature selection is not what makes the segment-wise scores high. (DS-2 keeps all 171 features, so ranking cannot matter there.)

6.3 Raw-channel baseline: is SMVMD needed?

The same 9 features computed directly on each preprocessed channel, with no SMVMD:

Setting	Raw: segment-wise	Raw: subject-wise	SMVMD: subject-wise	Raw: children correct
DS-1 Rest	100.00%	66.15%	62.73%	9/14
DS-1 Music	99.22%	67.55%	43.01%	8/14
DS-1 Rest+Music	100.00%	67.62%	62.34%	9/14
DS-2 ADHD	87.49%	57.23%	57.54%	69/121

6.4 What the features encode

Setting	Silhouette by class	Silhouette by child	Child identified (1-NN)	Chance
DS-1 Rest	0.075	0.218	98.91%	7.14%
DS-1 Music	0.066	0.193	99.07%	7.14%
DS-1 Rest+Music	0.067	0.173	99.30%	7.14%
DS-2 ADHD	0.026	-0.150	65.39%	0.83%

Silhouette: how well points group by a label (higher = tighter groups). The 1-NN probe predicts *which child* produced a window, using segment-wise 10-fold CV.

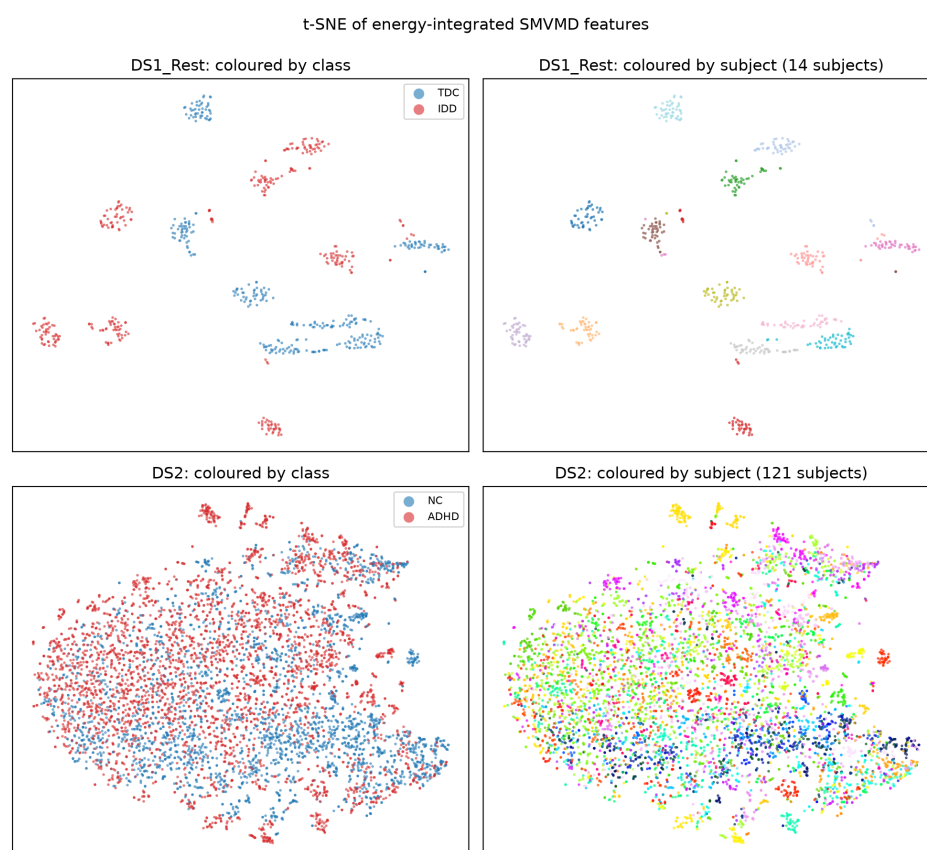


Figure 1. t-SNE of the SMVMD features. Left: coloured by class; right: coloured by child. Top: DS-1 rest (one island per child); bottom: DS-2.

6.5 Accuracy per child

DS-1 rest (leave-one-subject-out): 4 children are right in every window and 3 are wrong in every window — the model decides per child. DS-2: accuracy varies continuously; 46 of 121 children have fewer than half of their windows right.

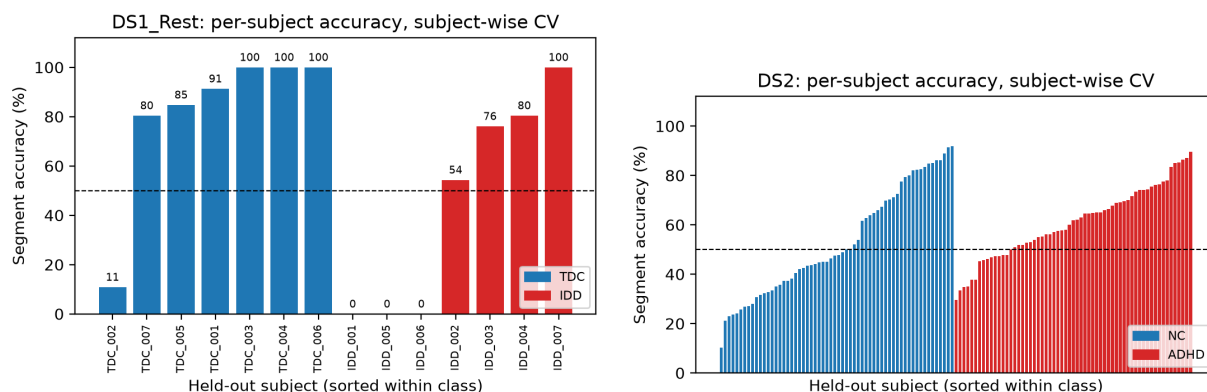


Figure 2. Fraction of each held-out child's windows classified correctly. Left: DS-1 rest; right: DS-2. Dashed line: 50%.

6.6 Number of SMVMD modes

DS-1: mean K 10.78 (IDD 11.37, TDC 10.19); 5.7% of windows hit the 20-mode safety cap. DS-2: mean K 4.55 (ADHD 4.74, NC 4.31); none hit the cap.

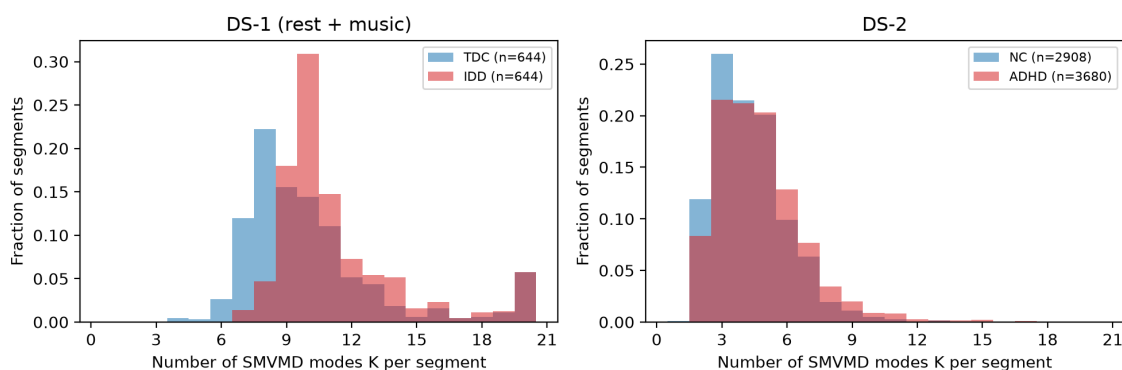


Figure 3. Number of SMVMD modes per window, by dataset and class.

7. What the results mean

The accuracy gap comes from recognising children. Each child's windows look very alike and very different from other children's. When windows from the same child are in both training and test sets, the classifier finds that child's other windows and copies the label. That works only for children already seen. For a new child — the real use case — accuracy is far lower.

Overlap adds to it on DS-2. Removing window overlap lowers DS-2 segment-wise accuracy from 85.81% to 74.86% while subject-wise stays at about the same level (57.54% → 58.04%): shared samples between overlapping windows were inflating the score. On DS-1 the child signature is strong enough without overlap. Part of the DS-2 drop may also come from having half as many windows.

SMVMD has not yet shown a benefit. Under subject-wise evaluation the raw-channel features beat the SMVMD features on all three DS-1 settings and are about equal on DS-2, even though the KNN settings were tuned by the paper for SMVMD features.

This is about evaluation, not about one paper. Splitting windows rather than children is common in EEG studies. The subject-wise numbers are the honest estimate for screening new children.

8. Caveats and things to be ready for

8.1 Limitations of the current work

- KNN hyperparameters come from the paper's Table II; its Bayesian optimisation was not re-run, and they were not re-tuned inside folds (planned: nested CV).

- DS-1 uses the dataset's cleaned recordings instead of re-running the paper's EEGLAB ICA/ADJUST cleaning (MATLAB-only).
- The paper does not say how SMVMD decides the number of modes. Here: stop when residual energy < 1% of the window, with a 20-mode safety cap (5.7% of DS-1 windows hit it).
- The SMVMD implementation is a simplified successive ADMM; SMVMD reconstruction is deliberately incomplete because $\tau = 0$, as in the paper.
- With only 14 children in DS-1, each child is 7.14% of the data, so DS-1 subject-wise numbers are coarse.
- DS-2 grouped folds are not stratified by class; 3 children tied 50/50 in the vote (counted as NC).
- The t-SNE is a picture, not a proof; the silhouette and 1-NN numbers are the quantitative evidence. UMAP was not run (not installed).
- Segment-wise “children correct” is always perfect because every child is in training; it is not a meaningful result.

8.2 Likely viva questions

Question	Short answer
Why is your accuracy lower than the paper's?	The paper reports 100% on all three DS-1 settings and 99.17% on DS-2. Under the same segment-wise protocol this reproduction gets 96.43%–99.77% on DS-1 and 85.81% on DS-2. Possible reasons: the unspecified SMVMD stopping rule, the simplified SMVMD, DS-1 cleaning, and hyperparameters not re-optimised. The MSE report shows own results without comparison.
Is subject-wise evaluation not too pessimistic?	It matches the real use: a new child is never in the training set. Segment-wise answers a different question (known children).
Could the drop be because of less training data?	LOSO trains on 13 of 14 children, nearly all data; the drop is too large for that, and the child-ID probe shows the cause.
Is ADHD detection valid on DS-2?	Medication is perfectly aligned with the label, so the dataset cannot separate ADHD from methylphenidate effects.
Why not use deep learning?	The same leakage would affect any model; fixing the evaluation comes first. Model changes are planned inside the nested protocol.

9. Mid-semester report status

- LaTeX report with the IIT Indore title page (logo), compact body, six required sections, three references (the paper and the two datasets). Upload report/mse_report_overleaf.zip to Overleaf and compile main.tex.
- Every number in the report is a macro generated from results_summary.json by report/build_tables.py, so none is typed by hand; report/check_report.py verifies macros, citations and environments.
- Written from scratch; the only 40+ letter overlap with the paper's text is the phrase “attention-deficit hyperactivity disorder (ADHD)”.
- To do before submission: fill the title-page placeholders (name, roll number, discipline, supervisor, department) and confirm on Overleaf that the body is at most 7 pages.

10. Plan until the final evaluation

Period	Work	Purpose / output
August – 25 Sep	Done: reproduction on both datasets; subject-wise protocols, overlap, raw-channel and embedding analyses; MSE report. Mid-evaluation on 25 Sep.	This record.
28 Sep – 11 Oct	Nested subject-wise cross-validation: tune KNN settings and feature count inside each outer fold (Bayesian search); bootstrap confidence intervals over children.	An accuracy estimate for new children with no leakage of any kind, with error bars.
12 – 25 Oct	Ablation ladder under the nested protocol: raw channels → fixed frequency bands → VMD → MVMD → SMVMD; sensitivity to the SMVMD stopping rule and mode cap.	Shows which step, if any, adds real value.
26 Oct – 8 Nov	Interpretability: permutation importance and SHAP on subject-wise models; channel-level scalp maps; theta/beta power ratio as a simple ADHD baseline.	What the model relies on, and whether it is physiologically plausible.
9 – 15 Nov	Cross-dataset feature analysis on the 10 shared electrodes (F7, F3, F4, F8, T7, T8, P7, P8, O1, O2; T3/T4/T5/T6 in DS-2 = T7/T8/P7/P8).	Whether patterns carry over between the two datasets.
16 – 22 Nov	Channel reduction under subject-wise evaluation.	How few electrodes are enough — relevant for cheap headsets.
23 – 30 Nov	Consolidate, re-run everything from scripts, final report; final evaluation and submission.	Final report and presentation.

11. How to re-run everything

From the project folder (C:\Users\Prateek\OneDrive\Desktop\btpnew):

```
python -m experiments.ds1_rest
python -m experiments.ds1_music
python -m experiments.ds1_rest_music
python -m experiments.ds2_adhd
python -m experiments.stage1_analysis
python report/build_tables.py
python report/check_report.py
python report/build_project_record.py
```

Reproductions reuse cached features when present (add --no-cache to recompute). Measured run times: DS-1 Rest 46 min, DS-1 Music 35 min, DS-1 Rest+Music 96 min, DS-2 ADHD 85 min (first run, 12-core machine).

12. Summary in simple words

What I have done

- I picked a 2025 research paper that uses one EEG method to detect two childhood disorders: IDD and ADHD.
- I rebuilt that whole method in Python and ran it on the same two public datasets (14 children for IDD, 121 for ADHD). While doing so I found and fixed several bugs, including one wrong feature formula.
- Tested the paper's way, my version gets 99.69% (IDD, rest) and 85.81% (ADHD).
- Then I tested it the way it would really be used: on children the model has never seen. Accuracy dropped to 62.73% (IDD, rest) and 57.54% (ADHD).
- I found out why: the EEG features mostly act like a fingerprint of each child. If the model has seen a child before, it recognises the child and copies the label. For a new child it has much less to go on.
- I also found that the complicated decomposition step (SMVMD) did not help compared with using the plain EEG channels, and that in the ADHD dataset all ADHD children were on medication, which the model cannot separate from ADHD itself.
- I wrote the mid-semester report with all of this, with every number filled in automatically from my results.

What I will do next

- Build a fair evaluation where even the tuning never sees the test children, and add error bars.
- Test step by step which parts of the method actually help (plain EEG, frequency bands, VMD, MVMD, SMVMD).
- Find out what the model looks at — which features and which brain regions — and compare with a simple, well-known ADHD measure (theta/beta ratio).
- Check whether patterns hold across both datasets using the 10 electrodes they share, and how few electrodes are enough.
- Write the final report and present it at the end of November.

In one sentence: I reproduced a published EEG method for detecting IDD and ADHD, showed that its very high accuracy mostly comes from recognising individual children rather than the disorder, and will spend the rest of the semester building and explaining a model that works on children it has never seen.